Marine Fish Quality Monitor: IoT Supervision of a Fishing Boat's Fish Hold

Ice temperature, meltwater clogging and spoilage-gas monitoring with a summary display, data logging and a phone dashboard for the boat owner

Category	Level	Main board
Internet of Things	High school	NodeMCU ESP8266

1. ABSTRACT

Fishing boats keep their catch in an insulated hold packed with crushed ice. Three things ruin the catch at sea: the ice running out, meltwater that cannot drain because the outlet is blocked so the fish soak in dirty water, and fish that start to rot (or have been treated with chemicals). The Marine Fish Quality Monitor watches all three with low-cost sensors on a NodeMCU ESP8266: a DHT11 for the hold temperature, an ultrasonic sensor for the height of the meltwater, a moisture probe that detects water reaching the fish, an MQ-3 for alcohol and chemical vapour and an MQ-5 for spoilage gas. It logs every reading, computes a freshness score, shows only a short summary on a 4-digit TM1637 display and flashes it with a fault code when anything goes abnormal. The boat owner joins the board's own WiFi and sees everything live on a phone or laptop. A built-in simulator plays a full voyage so the system can be demonstrated without sensors.

2. AIM OF THE PROJECT

- Keep the fish hold at ice temperature (0 °C) and warn when the ice is melting away.
- Measure the meltwater height and warn when too much water means the drain outlet is blocked.
- Detect alcohol / ammonia-type vapour from rotting fish or chemicals used on the catch.
- Log the data through the voyage, show a summary on the box and the full picture on the owner's phone or laptop.
- Provide a simulator so the whole system can be shown and tested without sensors.

3. PROBLEM STATEMENT AND NEED

Fish is one of the most perishable foods. On ice it stays fresh for days, but once the hold rises above 0 °C bacteria multiply quickly. Melted ice must drain away: if the drain clogs with scales and slime, the bottom layer of fish soaks in meltwater and spoils fast. Spoiling fish give off volatile compounds such as ammonia, amines and alcohols. The crew is busy fishing and the hold is closed, so these problems are often found only at the harbour, when the catch has already lost its price. Some sellers also use chemicals to make old fish look fresh. A cheap monitor that watches the hold all the time and tells the owner immediately protects both the catch and the buyer.

4. COMPONENTS USED AND COST

Component	Model	Qty	Purpose	Approx. cost
Controller	NodeMCU ESP8266 (ESP-12E)	1	Reads sensors, logic, logging, hotspot, web server	Rs. 280
Display	TM1637 4-digit 7-segment	1	Summary values; flashes fault codes	Rs. 90
Temperature sensor	DHT11	1	Hold (ice) temperature and humidity	Rs. 110
Ultrasonic sensor	HC-SR04	1	Meltwater height on the hold floor	Rs. 90
Moisture probe	Soil moisture probe + module	1	Water reaching the bottom fish	Rs. 80
Alcohol sensor	MQ-3 module	1	Alcohol / amine vapour: rot or chemicals	Rs. 150

Component	Model	Qty	Purpose	Approx. cost
Gas sensor	MQ-5 module	1	Spoilage (decomposition) gas trip	Rs. 150
Indicator	Red 5 mm LED + 220 Ω	1	Alarm light	Rs. 5
Resistors	1 kΩ, 2 kΩ, 4.7 kΩ	1 set	Echo divider, MQ-5 DO series resistor	Rs. 10
Power	5 V 1-2 A USB supply / power bank	1	Supply (MQ heaters need 5 V)	Rs. 150
Total				Rs. 1,115

Table 1: Bill of materials (approximate local prices)

5. CIRCUIT CONNECTIONS

The NodeMCU has only one analog input (A0). The MQ-3 uses it because its vapour level is shown as a number; the MQ-5 and the moisture probe use the digital outputs of their comparator modules, set with their potentiometers. The HC-SR04 echo is a 5 V signal, so it goes through a 1 kΩ / 2 kΩ divider. The MQ-5 DO is only pulled up by the module's 10 kΩ resistor, so a 4.7 kΩ series resistor is enough to keep the current into the pin tiny.

Part pin	NodeMCU pin	Notes
TM1637 CLK / DIO	D5 / D6	VCC from VIN (5 V)
DHT11 DATA	D4	VCC 3.3 V
HC-SR04 TRIG / ECHO	D7 / D8	ECHO through 1 kΩ / 2 kΩ divider, VCC 5 V
MQ-3 AO	A0	board has its own A0 divider, VCC 5 V
MQ-5 DO	D1	through 4.7 kΩ, VCC 5 V
Soil probe DO	D2	VCC 3.3 V
Red LED	D0	through 220 Ω to GND

Table 2: Wiring (full sheet in CONNECTIONS.md)

6. SYSTEM OVERVIEW

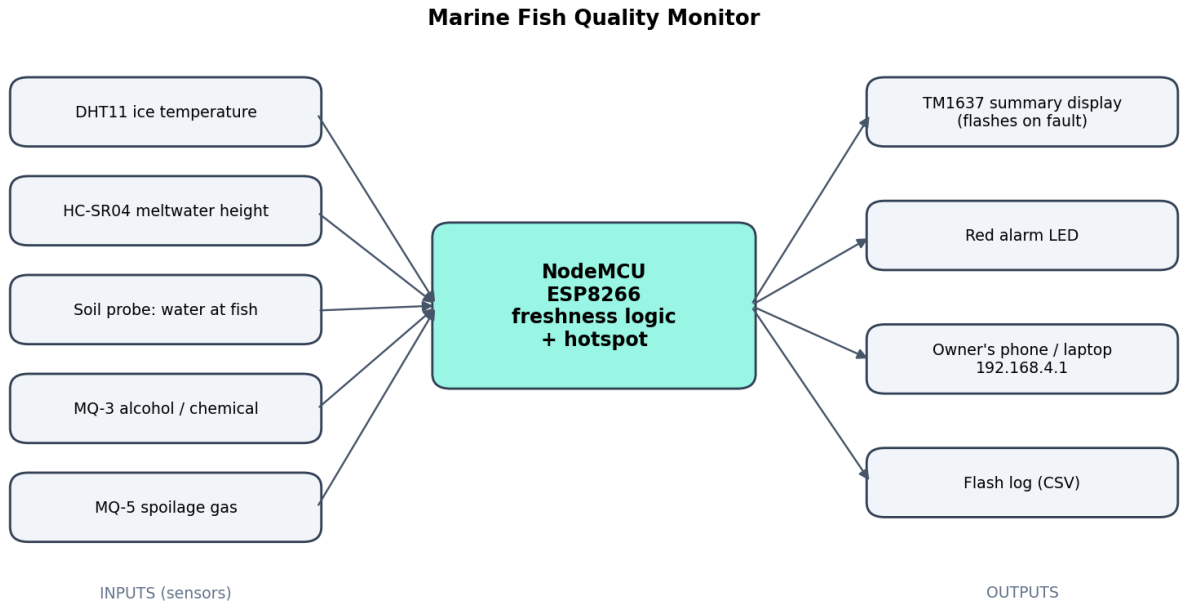


Figure 1: Block diagram

7. HOW I DID THIS (METHOD)

The ultrasonic sensor is fixed under the lid looking straight down. Meltwater height = (distance to the empty floor) - (measured distance); the median of the last five pings ignores splashes from the rolling boat. The

moisture probe is placed at the level of the lowest fish, so it confirms when the water has reached them. The DHT11 sits in the ice between the fish. The MQ-3 reading is averaged over 8 samples and the MQ gas alarms are ignored for the first 60 s while the heaters warm up.

A freshness score starts at 100 and loses points for temperature above 0 °C (6 per degree), meltwater above 2 cm (10 per cm), water at the fish (8), MQ-3 vapour above clean air and an MQ-5 trip (25). It is graded FRESH, GOOD, CHECK or SPOILED. The TM1637 normally cycles the temperature, water height, gas level and score. When anything is abnormal it flashes a code (ICE, drAn, ALCO, rOt) followed by the value, and the red LED blinks. Every minute a CSV row is written to the board's flash; the dashboard gives the board the real time from the phone so log rows get real dates. A missing sensor is shown as NO SENSOR, never as a made-up value.

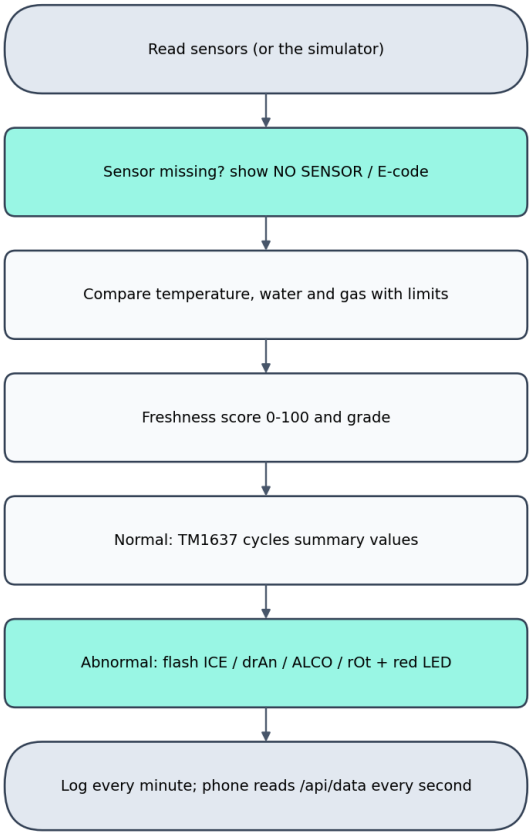


Figure 2: Program flow

8. SIMULATOR

Because the sensors are not always connected, the firmware has a simulator that feeds the same alarm, display, LED and logging code. In 'voyage story' mode it plays a 4-minute voyage: fresh catch, ice melting, drain outlet blocked, outlet cleared, fish rotting and fresh ice again. In 'manual' mode the owner moves sliders on the dashboard. The dashboard page also runs as an offline simulator on any laptop without the board.

9. TESTING AND RESULTS

The graph below is a replay of the firmware's own simulator and alarm formulas over one voyage story. Every alarm fires in the right phase and clears afterwards; the freshness score falls to 0 at the worst point (183 s, rotting fish without ice) and recovers to 97 with fresh ice.

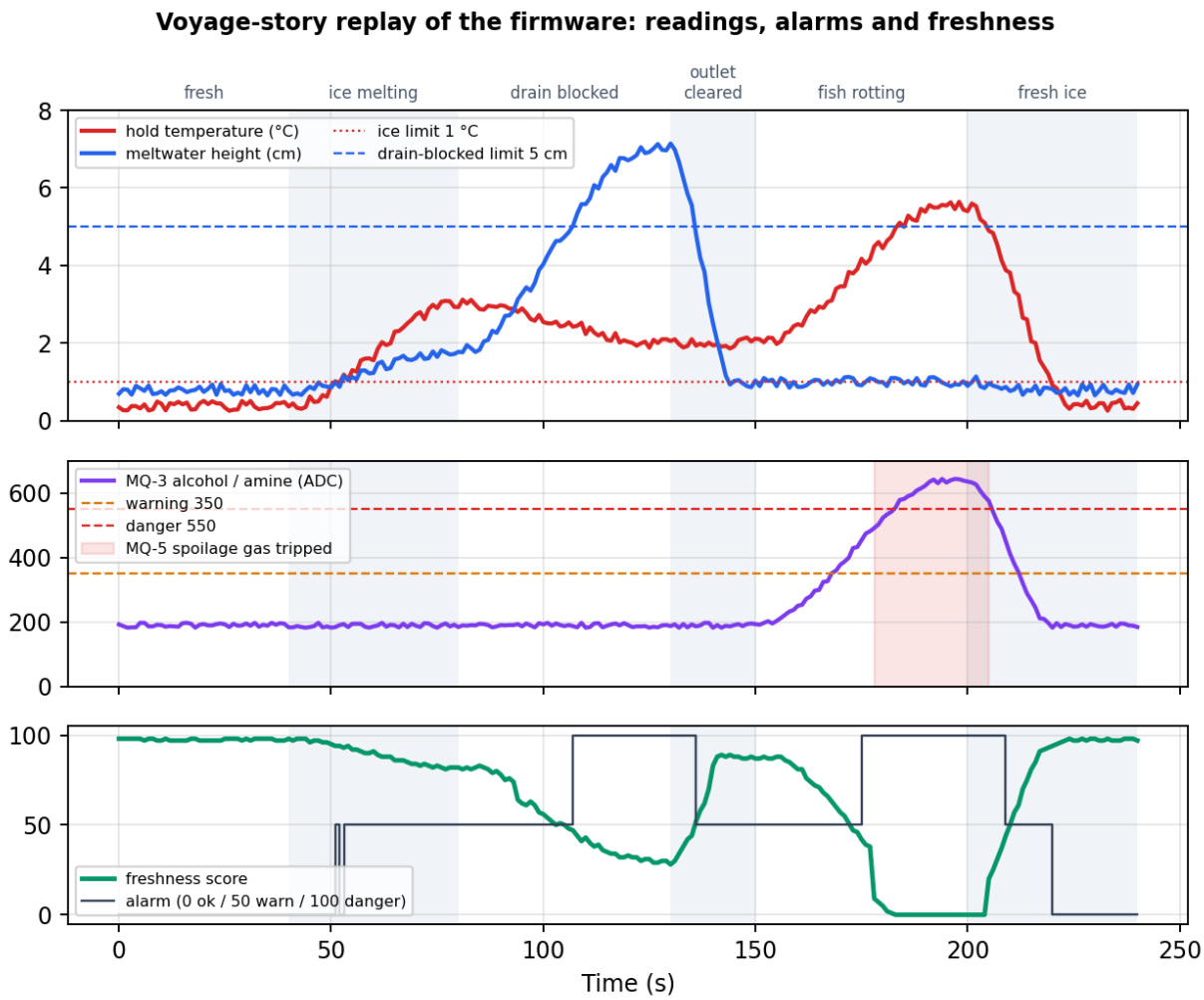


Figure 3: Readings, alarms and freshness during the simulated voyage

Phase of the voyage	What the sensors saw	Box display	First alarm
Fresh catch on ice (0-40 s)	0.4 °C, 0.8 cm water, MQ-3 ~190	cycles 0°C / H 1 / AL18 / Fr97	none
Ice melting (40-80 s)	temperature climbs to 3 °C	flashes ICE	51 s
Drain outlet blocked (80-130 s)	water rises to 7.1 cm, probe wet	flashes drAn	86 s
Outlet cleared (130-150 s)	water drains back to 1 cm	ICE only	-
Fish rotting (150-200 s)	5.5 °C, MQ-3 up to ~640, MQ-5 trips	flashes ALCO, rOt, ICE	168 s / 178 s
Fresh ice (200-240 s)	all values back to normal	summary again	cleared

Table 3: What the monitor reported in each phase

The firmware compiles for the NodeMCU (32 % of flash, 39 % of RAM) and the dashboard was tested in a phone-sized browser in every mode.

10. APPLICATIONS

- Fishing boats on multi-day voyages.
- Insulated fish boxes at harbours and auction halls.
- Fish transport vans and cold storage rooms.
- Fish markets checking for chemically treated fish.

11. ADVANTAGES AND LIMITATIONS

- Watches ice, meltwater and spoilage together for about Rs. 1,100.
- Owner sees it on any phone; no internet or app needed.
- Logs the whole voyage; simulator for demonstration and training.
- Limitation: the DHT11 cannot read below 0 °C; a DHT22 or DS18B20 would be more precise.
- Limitation: MQ sensors give relative readings and must be calibrated in clean air; an MQ-135 or MQ-137 would measure ammonia more directly.

12. FUTURE IMPROVEMENTS

- Waterproof DS18B20 probe for exact ice temperature.
- GSM / LoRa link so the owner on shore gets alerts.
- Automatic drain pump when the meltwater is too high.
- Solar charging and an IP65 box for the deck.

13. CONCLUSION

The Marine Fish Quality Monitor turns a closed fish hold into something the owner can watch from a phone. It keeps a record of the voyage, raises a clear alarm when the ice runs out, the drain blocks or the fish start to spoil, and shows its readings honestly - helping fishermen land a fresher, more valuable catch.

14. REFERENCES

- FAO - Ice in fisheries (Fisheries Technical Paper 331).
- Hanwei Electronics - MQ-3 and MQ-5 datasheets.
- Aosong - DHT11 datasheet.
- ESP8266 Arduino core documentation (ESP8266WebServer, LittleFS).